

BS – 101 Engineering Physics

Lecture – 6

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Point Charge Inside a Spherical Surface

- Gauss's Law provides a different way to express the relationship between electric charge and electric field.
- It was formulated by a 18th century mathematician Carl Friedrich Gauss (1777-1855).

Point Charge Inside a Spherical Surface

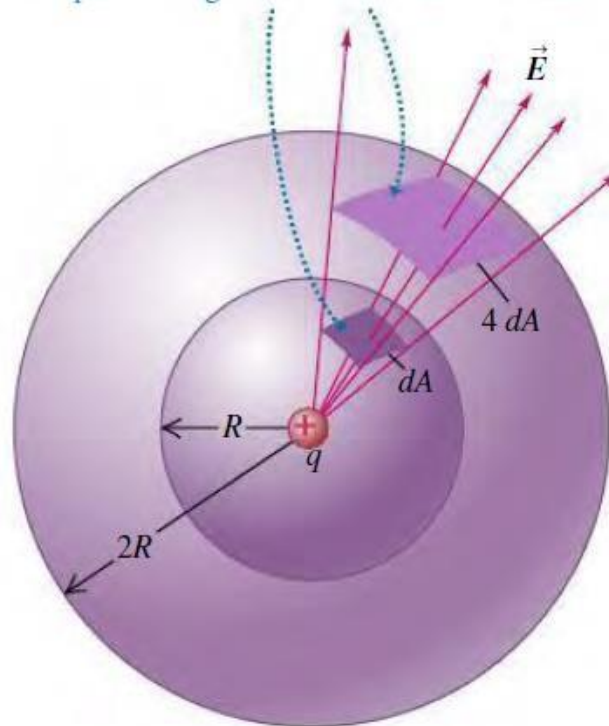
- Gauss's law states that the total electric flux through any closed surface (a surface enclosing a definite volume) is proportional to the total (net) electric charge inside the surface.

$$\Phi_E = EA = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2} (4\pi R^2) = \frac{q}{\epsilon_0}$$

Point Charge Inside a Spherical Surface

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The same number of field lines and the same flux pass through both of these area elements.



Various Charges inside an Irregular Surface

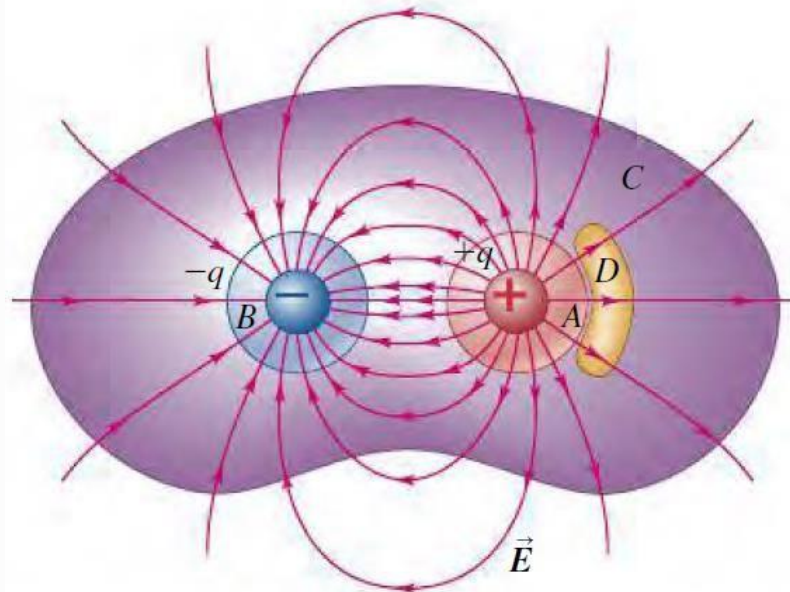
- $\varphi = \frac{Q_{encl}}{\epsilon_0}$

- Where

- $Q_{encl} = \text{Sum of all charges}$

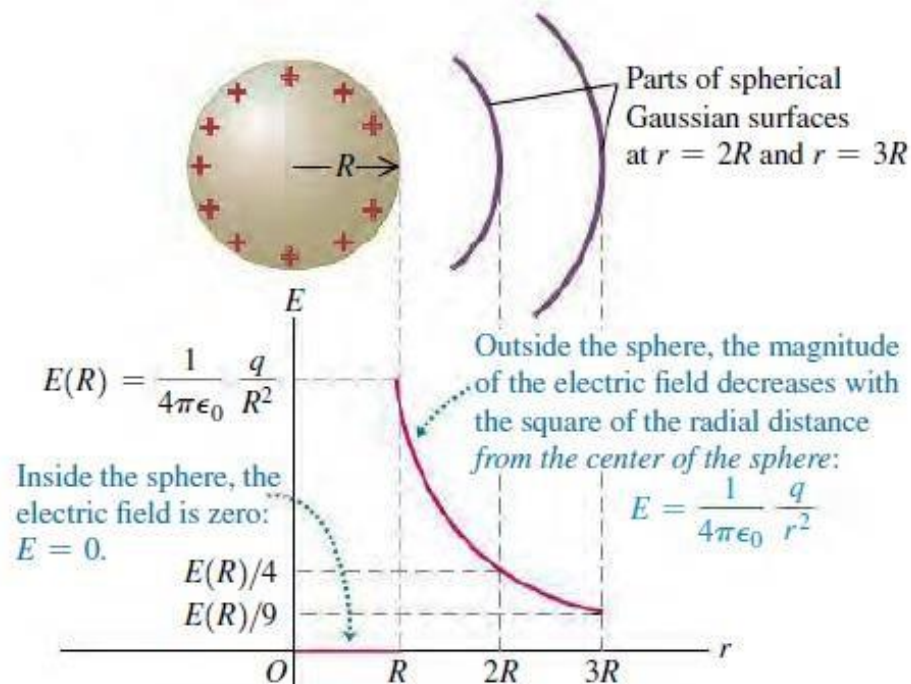
Various Charges inside an Irregular Surface

- Figure shows the field produced by two point charges $+q$ and $-q$ (an electric dipole). Find the electric flux through each of the closed surfaces A , B , C , and D



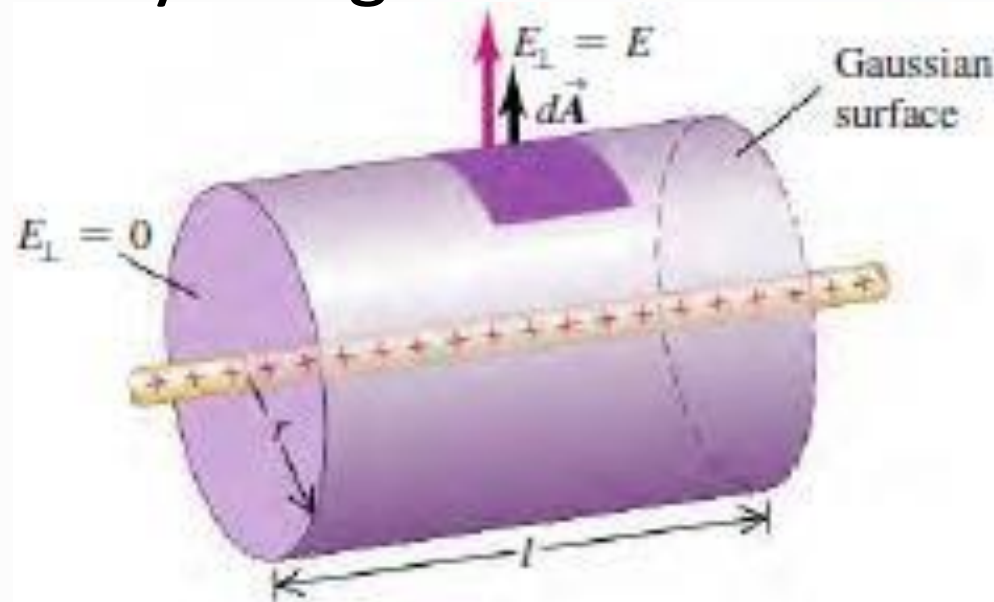
Field of a Charged Conducting Sphere

- We place a total positive charge q on a solid conducting sphere with radius R . Find $\mathbf{E}$ at any point inside or outside the sphere.



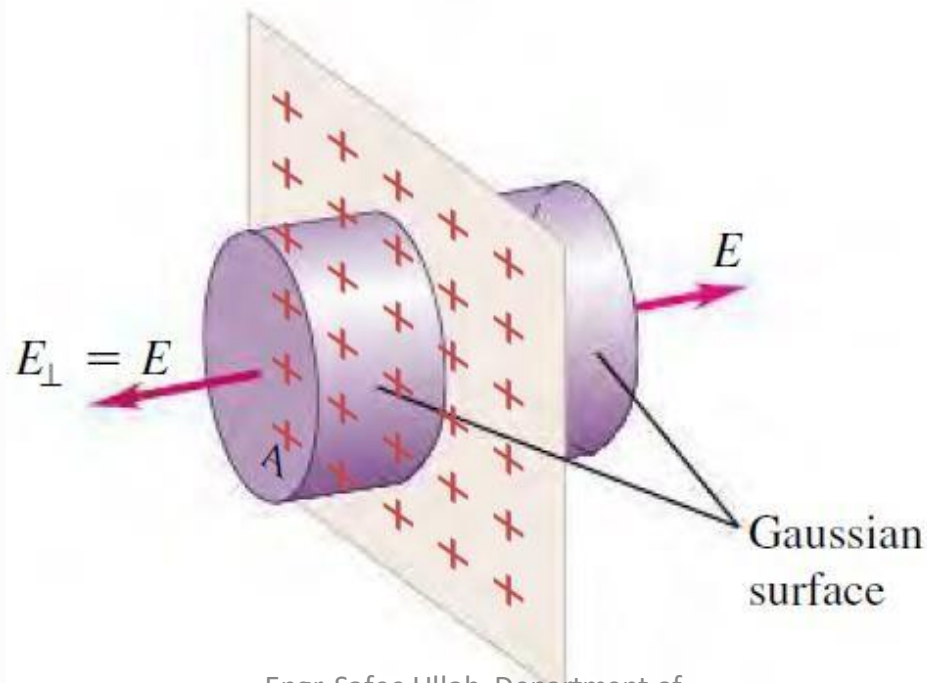
Field of a Uniform Line Charge

- Electric charge is distributed uniformly along an infinitely long, thin wire. The charge per unit length is λ (assumed positive). Find the electric field by using Gauss's law.



Field of an Infinite Plane Sheet of Charge

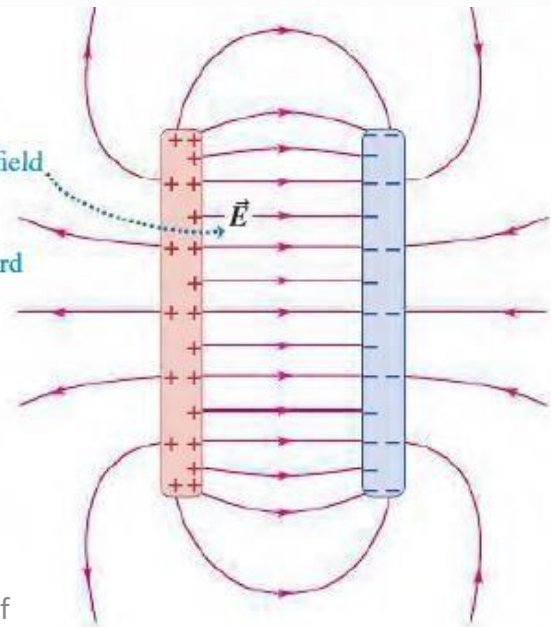
- Use Gauss's law to find the electric field caused by a thin, flat, infinite sheet with a uniform positive surface charge density σ .



Field Between Oppositely Charged Parallel Conducting Plates

- Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the surface charge densities are $+\sigma$ and $-\sigma$. Find the electric field in the region between the plates.

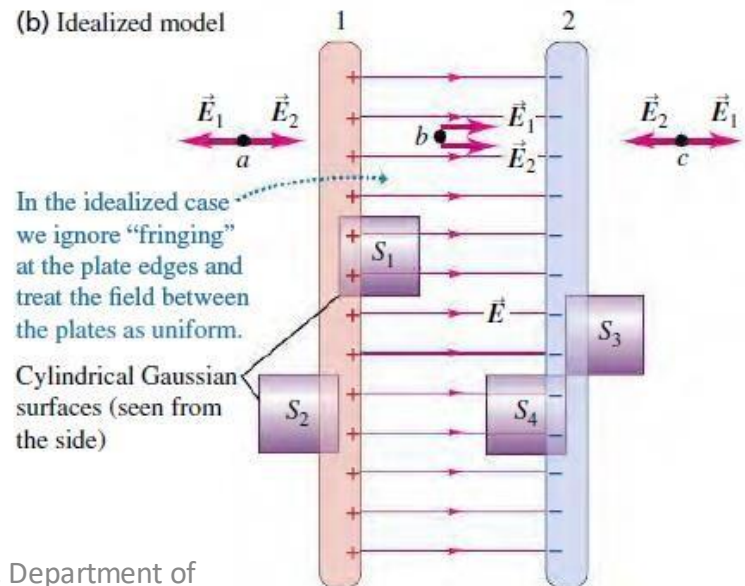
Between the two plates the electric field is nearly uniform, pointing from the positive plate toward the negative one.



Field Between Oppositely Charged Parallel Conducting Plates

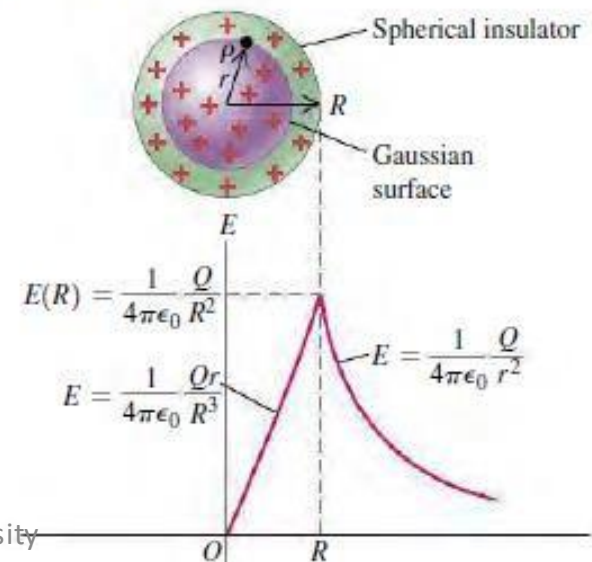
- Two large plane parallel conducting plates are given charges of equal magnitude and opposite sign; the surface charge densities are $+\sigma$ and $-\sigma$. Find the electric field in the region between the plates.

(b) Idealized model



Field of a Uniformly Charged Sphere

- Positive electric charge Q is distributed uniformly *throughout the volume* of an *insulating* sphere with radius R . Find the magnitude of the electric field at a point P a distance r from the center of the sphere.





Charge on a Hollow Sphere

- A thin-walled, hollow sphere of radius 0.250 m has an unknown charge distributed uniformly over its surface. At a distance of 0.300 m from the center of the sphere, the electric field points radially inward and has magnitude 1.80×10^2 N/C. How much charge is on the sphere?